

Empirical Evaluation of ProtoHedge: Interpretable Deep Hedging with Real Market Data

Yanis Zaim
Columbia University
New York, NY, USA
ymz2108@columbia.edu

Sayak Chakrabarty
Northwestern University
Evanston, IL, USA
sayak.chakrabarty@northwestern.edu

Imon Banerjee
Northwestern University
Evanston, IL, USA
imon.banerjee@northwestern.edu

ABSTRACT

Deep Hedging learns sequential hedging policies directly from market states, transaction costs, and a risk-sensitive objective, but its neural decisions are difficult to audit. ProtoHedge addresses this problem by routing each hedge through a finite set of representative market states, or prototypes: the action is determined by the current state’s similarity to those prototypes and their learned hedge responses. We test whether this performance–interpretability trade-off survives historical option data, where frictions, non-stationarity, and path dependence matter. ProtoHedge is compared with unhedged exposure, delta rules, and a common-protocol black-box Deep Hedging benchmark on European and arithmetic-average Asian short calls. The design uses chronological purged splits, one consistent listed hedge contract option per episode, cumulative-inventory accounting, payoff-sufficient Asian states, hard bounds, and validation selection. The outcome excludes the liability premium and therefore is not complete economic profit and loss. Across the current ten-underlying panel, validation-selected ProtoHedge improves the average mean and lower-tail offset relative to Deep Hedging for both liabilities. European policies also reduce average position-bound occupancy; the Asian bound result is weaker. Prototype use is concentrated, varies with market regimes, and can be traced to historical states. The evidence supports ProtoHedge as an inspectable empirical hedge without claiming uniform dominance across assets or objectives.

CCS CONCEPTS

• **Applied computing** → **Finance**; • **Computing methodologies** → *Instance-based learning; Neural networks; Sequential decision making.*

KEYWORDS

deep hedging, ProtoHedge, interpretable machine learning, option hedging, empirical finance

1 INTRODUCTION

Deep Hedging replaces a prescribed analytical hedge with a policy learned from market states, transaction costs, and a risk-sensitive objective. This flexibility is valuable when hedging is discrete, costly, constrained, or exposed to dynamics that depart from a stylized pricing model [1]. Its central practical weakness is auditability. Strong out-of-sample performance alone does not reveal which market states drive a hedge, whether comparable states receive comparable actions, or whether the policy has learned a stable economic rule rather than an opaque statistical association.

ProtoHedge addresses this limitation by routing each hedge decision through a finite set of representative market states, called prototypes. A current state is compared with these prototypes, and the resulting similarity weights combine prototype-associated hedge responses into the final action. This construction makes each action traceable to observed representative states while retaining the risk-sensitive learning objective of Deep Hedging. The original ProtoHedge study established architectural feasibility in Black–Scholes and stochastic-volatility simulations. It did not establish whether the same performance–interpretability tradeoff survives historical non-stationarity, changing liquidity and volatility, observed option paths, and path-dependent liabilities [2].

This paper addresses that empirical gap using historical U.S. equity and ETF option data. We preserve the original fixed-prototype architecture and compare it with transparent delta-style rules and a black-box Deep Hedging benchmark under a common state space, action space, objective, cost model, and position constraints. The evaluation uses chronological train, validation, and test periods; constructs rolling episodes only after the calendar is split; follows one listed hedge-option contract throughout each episode; selects every model configuration without test access; and evaluates uncertainty with paired block-bootstrap intervals. We study both a terminal European call and an arithmetic-average Asian call whose policy state includes the running payoff state.

Under the current empirical results, validation-selected ProtoHedge policies remain competitive with the Deep Hedging benchmark across both liabilities. On the European panel, separate mean- and tail-oriented selections improve the corresponding average liability-offset criteria while contacting hard position bounds less frequently on average. The Asian experiment preserves the mean and lower-tail competitiveness of ProtoHedge, although its position-bound advantage is weaker, showing that the tradeoff changes under path dependence. Prototype usage is concentrated, changes across return, volatility, and drawdown regimes, and can be traced to concrete historical states. These findings support a qualified conclusion: the fixed prototype bottleneck can preserve empirical hedging quality while making decisions more inspectable, but its advantage is not uniform across liabilities, assets, or evaluation criteria.

The paper makes four contributions:

- (1) We construct 35,987 contract-consistent 20-step episodes for ten U.S. equity and ETF underlyings using WRDS spot and option data.
- (2) We evaluate European and Asian short-call liabilities under inventory-based accounting, stylized transaction costs, hard position bounds, and a common-protocol comparison with classical and neural benchmarks.

- (3) We split by calendar, select configurations and checkpoints using validation data only, and apply paired circular-block inference.
- (4) We test interpretability through exact prototype-weighted action structure, prototype concentration, regime-dependent usage, and retrieval of nearest historical episodes.

2 PROTOHEDGE AND THE EMPIRICAL VALIDATION PROBLEM

2.1 Deep Hedging as Sequential Risk Management

Deep hedging replaces a prescribed replication formula with a policy learned from observed market states and a risk-sensitive hedging objective. Let $P_t \in \mathbb{R}^m$ be the vector of hedge-instrument prices, let h_t be the cumulative hedge inventory, and let τ_t denote time remaining. At time t , a policy observes a state such as

$$x_t = (P_t, h_{t-1}, \tau_t)$$

and outputs a trade increment

$$a_t = \pi_\theta(x_t), \quad h_t = \Pi_{\mathcal{H}}(h_{t-1} + a_t),$$

where $\Pi_{\mathcal{H}}$ enforces the admissible position set. Let $H(P_{0:T}) \leq 0$ be the terminal cash flow of the short liability and let κ_t be the per-unit transaction-cost vector. The premium-excluding liability offset can be written schematically as

$$O^{\pi_\theta} = H(P_{0:T}) + \sum_{t=0}^{T-1} h_t^\top (P_{t+1} - P_t) - \sum_{t=0}^{T-1} \kappa_t^\top |a_t|.$$

The agent is trained to maximize the optimized-certainty-equivalent monetary utility used in the original Deep Hedging framework. For the piecewise-linear utility selected in our experiments,

$$\mathcal{U}_\lambda(O) = \sup_{y \in \mathbb{R}} \mathbb{E}[(1 + \lambda) \min(O + y, 0) - y], \quad \lambda = 1.$$

The scalar y is learned jointly with the policy. We call this implementation setting the cvar monetary utility because that is its name in the original software; it must not be confused with the separately reported empirical CVaR_{5%}. The policy optimizes this risk-sensitive objective rather than reproducing a model-implied delta exactly. This formulation accommodates discrete rebalancing, costs, hard bounds, and dynamics that need not follow a stylized pricing model [1]. Because O excludes the initial premium received for selling the liability, it is not complete economic profit and loss; its purpose is to compare how effectively policies offset the same liability under a common objective.

A standard, or black-box, Deep Hedging policy represents π_θ by a feed-forward neural network. It is a necessary performance benchmark in this paper because it has the same information, objective, costs, and constraints as ProtoHedge. Its limitation is diagnostic rather than computational: an action can be evaluated without revealing which market states motivated it or whether the policy has learned an economically stable rule.

2.2 ProtoHedge: Fixed Prototypes as an Interpretable Policy

ProtoHedge retains the deep-hedging objective but restricts action generation to a finite collection of representative states

$$\mathcal{P} = \{p_1, \dots, p_K\}.$$

In the fixed-prototype design studied here, each p_k is an observed standardized state selected from the empirical state distribution. For a standardized current state $\tilde{x}_t$, nonnegative feature weights γ , and a shared learned projection f_ϕ , ProtoHedge forms

$$z_t = f_\phi(\gamma \odot \tilde{x}_t), \quad z_k^p = f_\phi(\gamma \odot p_k),$$

then measures squared distances and converts them into normalized prototype weights:

$$d_{t,k} = \|z_t - z_k^p\|_2^2, \quad \alpha_{t,k} = \frac{\exp(-d_{t,k})}{\sum_{j=1}^K \exp(-d_{t,j})}.$$

Each prototype has an associated trainable, bounded hedge response $b_k \in \mathbb{R}^m$. The action is the similarity-weighted combination

$$a_t = \sum_{k=1}^K \alpha_{t,k} b_k.$$

This decomposition is the central ProtoHedge mechanism. For every hedge decision, the model exposes the prototypes with the largest weights, their representative market states, and their associated actions. A reviewer can therefore trace a decision to examples of historically observed states rather than only to opaque network parameters. Concentrated use of a small subset of prototypes, systematic shifts in $\alpha_{t,k}$ across regimes, and retrieval of the nearest historical episodes provide direct, testable forms of interpretability [2].

The anchors p_k remain fixed during training in the present study; the projection f_ϕ and the prototype actions b_k are learned. This preserves the original ProtoHedge premise that a prototype should remain tied to an actual representative state, rather than drifting to an uninterpretable parameter value. Section 3 specifies how the prototypes are extracted, how K and similarity weighting are varied, and how the policy is trained.

2.3 Why the Empirical Test Matters

The original ProtoHedge study showed that prototype-based policies can remain competitive in Black–Scholes and stochastic volatility simulations [2]. Historical option data pose a materially different test: volatility and liquidity vary over time, option observations are noisy, trading is costly and constrained, and market paths contain regime shifts and discontinuities. The interpretability claim is also stronger on real data, because a useful prototype should correspond to a recognizable historical state rather than only to a simulator-generated configuration.

Accordingly, this paper evaluates whether ProtoHedge remains competitive with common-protocol Deep Hedging and transparent delta-style baselines while avoiding excessive reliance on hard position bounds. It also tests whether prototype use is concentrated, regime-dependent, and retrievable as concrete historical episodes. We conduct this evaluation for both terminal European short calls

and path-dependent Asian short calls, allowing the performance–interpretability tradeoff to be assessed under two economically distinct liabilities.

3 EXPERIMENTAL FRAMEWORK

This section describes the empirical design used to evaluate ProtoHedge on historical market data. Our implementation is deliberately conservative. We use fixed-length out-of-sample hedging episodes, explicit transaction costs, bounded trading actions, bounded cumulative hedge positions, validation-only model selection, and multi-seed evaluation. The goal is not merely to optimize a policy in sample, but to assess whether ProtoHedge remains competitive with Deep Hedging under a realistic and reproducible real-data protocol.

3.1 Data and Episode Construction

Our real-data experiments are based on historical U.S. equity and ETF option data obtained through WRDS, combining underlying spot-price histories with daily listed option records. The final panel contains ten liquid underlyings spanning broad equity indices, sector ETFs, fixed income, and single-name equities:

$$\{\text{AAPL, IWM, NVDA, QQQ, SPY, TLT, XLE, XLF, XLK, XLV}\}.$$

The available histories begin between January and February 2005, depending on the ticker, and end in December 2024.

The raw OptionMetrics chain is restricted to American-style calls with finite vendor Greeks and implied volatility, positive bids, non-crossed offers, and sufficient initial liquidity, and is aligned with the corresponding CRSP daily spot series. Midquotes define the listed option price. We split the calendar before creating any rolling window: training observations end on December 31, 2018, validation observations run through December 31, 2021, and test observations begin in 2022. A 19-observation embargo, equal to the overlap of adjacent 20-step windows, is removed after each boundary. Windows are then generated independently inside each period, so no date can occur in more than one of the training, validation, and test sets.

For every eligible 20-step spot window, we select the listed hedge option on the first date according to:

- (1) absolute distance between strike and spot,
- (2) absolute distance between the contract’s days-to-expiration and a 30-day target maturity,
- (3) bid–ask spread,
- (4) open interest (descending), and
- (5) volume (descending),

subject to an initial maturity filter of 20 to 40 calendar days, positive open interest, relative bid–ask spread no greater than one, and absolute moneyness no greater than 10%. Crucially, we retain an episode only if the same optionid, expiration, and strike remain valid on every episode date and the contract expires after the liability horizon. Thus

$$C_{i,t} = C(j_i, t), \quad t = 0, \dots, T,$$

for one fixed listed contract j_i per episode. The option hedge is distinct from the synthetic European or Asian liability, and no unmodeled option roll occurs inside an episode.

Ticker	Train	Validation	Test
AAPL	2,230	693	714
IWM	2,501	701	715
NVDA	1,944	678	659
QQQ	2,491	716	695
SPY	2,531	718	715
TLT	2,156	713	714
XLE	2,369	655	713
XLF	2,100	691	713
XLK	1,781	704	707
XLV	1,893	686	691
Total	21,996	6,955	7,036

Table 1: Contract-consistent 20-step episodes after the chronological split and 19-observation boundary embargo. Windows may overlap within a period but never across train, validation, and test periods.

The resulting per-date feature vector is

$$x_t^{\text{raw}} = (S_t, C_t, \Delta_t^{\text{call}}, v_t, \sigma_t^{\text{impl}}),$$

where S_t is the underlying spot, C_t is the selected call price, Δ_t^{call} is the selected call delta, v_t is the selected call vega, and σ_t^{impl} is implied volatility. These features are stored in the episode tensor in the fixed order

$$(\text{spot}, \text{call_price}, \text{call_delta}, \text{call_vega}, \text{ivol}).$$

Historical hedging episodes are constructed as rolling 20-step windows within each chronological period. A new block begins whenever consecutive observations are more than four calendar days apart, preventing windows from joining disjoint histories. Letting $\tau = 20$ denote the episode length, each sample is a tensor in

$$\mathbb{R}^{\tau \times 5}.$$

Across the ten-ticker panel, the corrected contract-consistent procedure yields 35,987 episodes: 21,996 training, 6,955 validation, and 7,036 test episodes.

Although the raw episode tensor contains five market features, the European policies consume the state used by the Deep Hedging gym:

$$(\text{price}, \text{delta}, \text{time_left}),$$

where price contains current spot and the price of the fixed listed call, delta is the current cumulative inventory in both hedge instruments, and time_left is time to the liability horizon. For the Asian liability, the policy additionally observes running-average moneyness,

$$m_{i,t}^{\text{Asian}} = \frac{\bar{S}_{i,t}}{K_i}, \quad \bar{S}_{i,t} = \frac{1}{t+1} \sum_{u=0}^t S_{i,u}.$$

The same feature is included in Deep Hedging, ProtoHedge, prototype clustering, similarity calculation, and prototype interpretation. This prevents two paths with identical current prices and inventories but different accumulated averages from being treated as the same liability state.

Finally, for numerical stability and cross-path comparability, the real-data world normalizes spot, call price, and call vega by

each path's initial spot level. Under this normalization, a liability struck at-the-money at inception has normalized strike one, while remaining equivalent to a strike of S_0 in raw price units.

3.2 Hedging Tasks

We study two liabilities on the same historical episode panel.

European ATM short call. The primary task is to hedge a short at-the-money European call over a 20-step historical episode. If $S_{i,0}$ denotes the initial spot on episode i , then the strike is set to

$$K_i = S_{i,0}.$$

The liability payoff at maturity is

$$H_i^{\text{Euro}} = -\max(S_{i,T} - K_i, 0),$$

where the negative sign reflects the fact that the hedger is short the option.

Asian ATM short call. The path-dependent extension replaces the European liability with an arithmetic-average Asian call, again struck at-the-money at episode start:

$$K_i = S_{i,0}.$$

Let the episode contain T trading decisions. The Asian liability is

$$H_i^{\text{Asian}} = -\max\left(\frac{1}{T} \sum_{t=1}^T S_{i,t} - K_i, 0\right).$$

In the implementation used for the final paper sweep, the averaging window spans the full historical episode.

Hedge instruments and actions. The hedge may trade in two liquid instruments:

- (1) the underlying spot instrument, and
- (2) the selected near-ATM listed call option.

The action at each step is therefore two-dimensional,

$$a_t = (a_t^{(S)}, a_t^{(C)}),$$

where each component denotes a trade increment rather than an absolute target position. The cumulative hedge inventory evolves recursively as

$$\delta_t = \delta_{t-1} + a_t.$$

Step-wise hedge accounting. All reported experiments use step-wise inventory accounting. If $\delta_{i,t}^{(S)}$ and $\delta_{i,t}^{(C)}$ are the cumulative positions held after trading at t , and $\Delta S_{i,t} = S_{i,t+1} - S_{i,t}$ and $\Delta C_{i,t} = C(j_i, t+1) - C(j_i, t)$ are one-step changes, then the trading gain is

$$\text{TG}_i = \sum_{t=0}^{T-1} \left(\delta_{i,t}^{(S)} \Delta S_{i,t} + \delta_{i,t}^{(C)} \Delta C_{i,t} \right).$$

The use of cumulative inventory is essential because the real-data world provides one-step price changes. It is algebraically equivalent to the return-to-terminal accounting in the original synthetic ProtoHedge world, but it does not incorrectly discard positions established at earlier dates.

Transaction costs. We impose the stylized transaction-cost specification inherited from the experimental framework separately on spot and option trades. It is a deterministic friction proxy rather than reconstructed executable bid-ask slippage. The per-unit spot trading-cost coefficient is

$$c_{i,t}^{(S)} = 0.0002 |S_{i,t}|.$$

The per-unit option trading-cost coefficient is

$$c_{i,t}^{(C)} = 0.02 |v_{i,t}| + 0.0002 |\Delta_{i,t}^{\text{call}}| + 0.0005 |C_{i,t}|.$$

The realized trading cost on episode i is therefore

$$\text{Cost}_i = \sum_{t=0}^{T-1} \left(|a_{i,t}^{(S)}| c_{i,t}^{(S)} + |a_{i,t}^{(C)}| c_{i,t}^{(C)} \right).$$

Liability offset and premium convention. For both liabilities, the empirical outcome is

$$O_i = H_i + \text{TG}_i - \text{Cost}_i.$$

The initial premium of the synthetic liability is not observed. In particular, the listed call used as a hedge is a distinct contract and its initial quote is not the premium of the European or Asian liability. We therefore do not add that quote to O_i and do not call O_i economic profit and loss. Higher values instead mean that the trading strategy offsets more of the common short liability after costs. A complete terminal P&L would require an independently specified liability premium $V_{i,0}$:

$$\text{P\&L}_i^{\text{terminal}} = V_{i,0} + O_i.$$

All model comparisons in this paper use O_i consistently.

3.3 Models and Baselines

We compare ProtoHedge against both classical baselines and a black-box neural benchmark.

Unhedged. This baseline takes no hedge action:

$$a_t \equiv 0.$$

Its liability offset is simply the liability payoff.

Spot-Delta. This is a rule-based baseline that hedges only with the underlying spot instrument. At each step, the target cumulative spot position is set equal to the observed delta of the selected near-ATM call. The option hedge position remains zero. This baseline uses the market-provided option delta directly and therefore provides a transparent classical reference point.

Spot-Delta Band. This baseline extends Spot-Delta with a no-trade band. Instead of rebalancing to the point target at every step, the current cumulative spot position is left unchanged as long as it remains within

$$[\Delta_t^{\text{call}} - b, \Delta_t^{\text{call}} + b],$$

where $b \geq 0$ is a band width. If the current position exits the band, it is projected back to the closest boundary. We tune b on the validation set using the grid

$$b \in \{0.00, 0.01, 0.02, 0.05, 0.10, 0.15\},$$

selecting the band that maximizes validation mean liability offset. This provides a friction-aware transparent comparator in the spirit of classical no-transaction-region hedging [7].

Deep Hedging benchmark. The main black-box benchmark is a feed-forward Deep Hedging policy trained end-to-end under the same world, friction model, and objective as ProtoHedge. For the European task, the agent uses

$$(\text{price}, \text{delta}, \text{time_left}),$$

and outputs a two-dimensional trade increment. The Asian version adds m_t^{Asian} . In the panel runs, the benchmark network has width 20, depth 3, and Softplus activations.

ProtoHedge. ProtoHedge uses the same task-specific state representation and output space as the Deep Hedging benchmark, but constrains action generation through a prototype layer. In the full grid, we vary:

- (1) the number of prototypes

$$K \in \{10, 25, 50, 100\},$$
- (2) the prototype source policy, using either the trained Deep Hedging benchmark or the Spot-Delta baseline as the source from which prototypes are extracted, and
- (3) whether similarity is weighted or unweighted.

Prototype extraction follows the same reusable pipeline throughout the repo. For a given source policy, we:

- (1) construct the flattened feature matrix seen by the policy,
- (2) standardize it with a StandardScaler,
- (3) apply K -means clustering in standardized feature space [4], and
- (4) store, for each cluster, the nearest real standardized feature vector rather than the raw cluster centroid.

This last step matters because it keeps each stored prototype tied to an actually observed market state rather than to an abstract average point with no direct empirical counterpart.

The shared projection f_ϕ is a two-layer map

$$D \longrightarrow 32 \longrightarrow D$$

with a ReLU between layers. Similarity uses unit temperature. In the unweighted specification, all feature-distance weights equal one. In the weighted specification, time remaining receives weight five and all other state components receive weight one. These weights and the prototypes remain fixed; the projection and bounded prototype actions are trainable. The action bounds are implemented with the same smooth soft-clipping logic as the original ProtoHedge layer. This keeps the present paper focused on empirical validation of the original fixed-anchor design rather than the separate question of moving prototypes.

3.4 Training and Model Selection Protocol

All real-data neural models are trained under the same multi-seed protocol.

Train/validation/test splits. The split is fixed by calendar and is identical across random seeds:

training: through 2018-12-31,
 validation: 2019 through 2021,
 test: 2022 through 2024.

The calendar is split before windows are constructed, and the 19-observation embargo at each boundary prevents a 20-step window

from crossing periods. Random seeds therefore affect neural initialization and prototype clustering, not test membership. The main panel uses the three seeds

$$(1234, 2345, 3456).$$

Optimization. All neural models are trained for 800 epochs using the same optimized-certainty-equivalent cvar monetary utility described in Section 2.1, with risk-aversion parameter one. This training objective is distinct from the empirical 5% tail metric used for evaluation. The optimizer uses learning rate

$$10^{-3},$$

together with clipping safeguards already built into the training pipeline. Classical baselines are not trained in the neural sense, but they are evaluated on exactly the same splits and under the same friction model.

Feature normalization and state semantics. The real-data world is run in normalized mode. Spot, call price, and call vega are divided pathwise by the initial spot. European policies observe current hedgeable prices, current hedge inventory, and time remaining; Asian policies also observe running-average moneyness. This keeps the neural state comparable across underlyings while supplying the payoff-relevant state of each liability.

Validation-based checkpoint selection. Model selection is based on the validation split, not on the training loss alone. Let L_{val} denote the validation objective. For the final robust panel runs, checkpoints are ranked by the selection score

$$S = L_{\text{val}} + \alpha_a \overline{|a|} + \alpha_\delta \overline{|\delta|} + \alpha_b \text{BO} + \alpha_p \text{PT},$$

where $\overline{|a|}$ is mean absolute action magnitude on the validation set, $\overline{|\delta|}$ is mean absolute cumulative position magnitude, BO is validation bound occupancy, and PT is the fraction of validation paths that touch any cumulative bound. In the reported full runs, we use

$$\alpha_a = 0.005, \quad \alpha_\delta = 0.010, \quad \alpha_b = 0.100, \quad \alpha_p = 0.$$

Thus, validation selection explicitly favors models that do not rely excessively on large actions, large cumulative positions, or persistent boundary saturation.

Validation-only configuration selection. Checkpoint selection and architecture selection are separate. First, every candidate is trained on the training period, and its checkpoint is chosen by the validation score above. Second, candidate-level validation metrics are averaged across the three seeds. We apply the prespecified validation bound screen and freeze two prespecified selections:

- (1) *Validation-Mean:* maximize validation mean liability offset, with validation CVaR, lower bound occupancy, smaller K , and model name as deterministic tie-breakers;
- (2) *Validation-Tail:* maximize validation CVaR_{5%} of liability offset, with validation mean, lower bound occupancy, smaller K , and model name as tie-breakers.

This selection covers K , weighted versus unweighted similarity, and prototype-source policy. Regularization coefficients, action bounds, position bounds, selection objectives, and the validation robustness

Component	Prespecified value
Episode length	20 observations
Train / validation / test	2005–18 / 2019–21 / 2022–24
Boundary embargo	19 observations
Random seeds	1234, 2345, 3456
Neural epochs	800
Learning rate	10^{-3}
Deep Hedging network	width 20, depth 3, Softplus
Prototype counts	10, 25, 50, 100
Prototype sources	Deep Hedging, Spot-Delta
Similarity	weighted, unweighted
Trade / position bounds	$[-1, 1]$ per instrument
Validation occupancy screen	$BO \leq 0.50$

Table 2: Common experimental protocol. Hyperparameter and checkpoint choices are made on training and validation data; the test calendar is fixed across seeds.

threshold are prespecified before the test stage. Candidate ProtoHedge models are not evaluated on test. After both selectors are written to disk, the baselines, the validation-selected Deep Hedging checkpoint, and only the frozen ProtoHedge configurations are evaluated once on the test period.

Regularization. In addition to validation-time selection penalties, the training loss includes mild regularization terms on action and cumulative-position magnitude:

$$\lambda_a = 0.001, \quad \lambda_\delta = 0.002.$$

Concretely, if L_{hedge} is the base objective, then the optimized loss is

$$L_{\text{train}} = L_{\text{hedge}} + \lambda_a \mathbb{E}[|a|] + \lambda_\delta \mathbb{E}[|\delta|].$$

These penalties are deliberately mild: they do not dominate the objective, but they discourage the model from achieving performance through unnecessarily aggressive trading.

Trade bounds and cumulative position bounds. The final reported real-data runs impose both action-level and cumulative-position bounds. At every step and for each traded instrument, trade increments are clipped to

$$[-1, 1],$$

and cumulative hedge positions are also constrained to

$$[-1, 1].$$

These bounds are scientifically important. In exploratory unconstrained runs, apparent performance improvements could be accompanied by heavy saturation at the position boundaries. The bounded protocol is designed to test whether ProtoHedge remains competitive without relying on unrealistic edge-of-feasible-region behavior.

Paper sweep design. For each ticker and liability, the full sweep evaluates the complete ProtoHedge grid across all three seeds on training and validation data. Selection is performed separately by ticker and liability using the fixed rules above. Only the resulting validation-selected configurations enter that ticker’s test comparison.

3.5 Evaluation Metrics

We evaluate models at both the hedging-performance level and the interpretability level.

Primary performance metrics. For each fitted model and each test split, we compute:

- (1) mean liability offset,

$$\mathbb{E}[O],$$

- (2) empirical lower-tail Value-at-Risk and Conditional Value-at-Risk,

$$\text{VaR}_{5\%}(O), \quad \text{CVaR}_{5\%}(O),$$

- (3) root mean squared liability offset,

$$\text{RMSE}(O) = \sqrt{\mathbb{E}[O^2]},$$

- (4) downside deviation,

$$\text{DD}(O) = \sqrt{\mathbb{E}[\min(O, 0)^2]},$$

and realized mean trading-cost and trading-gain summaries. Higher mean and CVaR are better; lower RMSE and downside deviation are better. We use the lower-tail empirical expected-shortfall convention for gains [6]. We record the negative-offset rate only as an implementation diagnostic. We do not present it as shortfall probability or a headline economic metric: because the premium is excluded, zero is not a break-even P&L threshold, and an unhedged out-of-the-money liability mechanically produces many zeros while a hedge incurs trading costs.

Position-bound diagnostics. Because constrained hedging behavior is a central part of the empirical question, we also record:

- (1) the fraction of all path-step-instrument positions lying at either cumulative position bound,
- (2) the fraction lying specifically at the upper or lower bound, and
- (3) the fraction of paths that touch any cumulative position bound at least once.

We refer to the first quantity as *bound occupancy*. Lower bound occupancy is interpreted as evidence that a model is operating more consistently in the interior of the feasible region, which is economically preferable all else equal.

Robust-screen criterion. The prespecified screen is calculated on validation data only. A candidate passes when average validation bound occupancy is no greater than 0.50; the path-touch threshold is recorded but not binding in the reported setup. Test bound occupancy remains a diagnostic outcome and cannot determine whether a model enters the paper comparison.

Dependence-aware uncertainty. Adjacent episodes within a period may share 19 of 20 dates, so individual windows are not treated as independent observations. For each validation-selected ProtoHedge policy and its matched Deep Hedging benchmark, we first average the pathwise test offsets across seeds and then construct paired 95% circular-block bootstrap intervals with block length 20 and 2,000 replicates. We report intervals for differences in mean offset and CVaR_{5%} and for improvements in RMSE and downside deviation; all are oriented so positive values favor ProtoHedge.

Panel intervals independently block-resample within ticker, calculate each ticker statistic, and average with equal ticker weight [5].

Prototype concentration metrics. For ProtoHedge models, we further compute:

- (1) average prototype weight across all paths and time steps,
- (2) top-1 prototype frequency, i.e. the proportion of decisions for which each prototype receives the highest weight, and
- (3) cumulative usage mass of the top- k prototypes.

These metrics quantify whether decisions are routed through a small number of dominant prototype states or diffusely across many prototypes.

Regime-sensitive interpretability metrics. To test whether prototype use changes systematically with market conditions, we form episode-level regime labels from the historical paths:

- *return regime*, using terciles of terminal return,
- *volatility regime*, using terciles of average implied volatility, and
- *drawdown regime*, using terciles of maximum realized draw-down.

This yields labels of the form

{down, middle, up},
 {low_vol, mid_vol, high_vol},
 {mild_dd, mid_dd, deep_dd}.

Within each regime, we summarize mean offset, tail risk, RMSE, downside deviation, and average prototype weights. This lets us test whether the same few prototypes dominate everywhere, or whether ProtoHedge reallocates attention across market states in a regime-sensitive way.

Nearest historical states. Finally, for the dominant prototypes, we recover the nearest historical feature vectors and corresponding underlying paths. This allows us to move beyond abstract prototype coordinates and ask whether the most-used prototypes correspond to recognizable empirical market scenarios. This step is central to the interpretability claim of the paper: the prototype mechanism is not merely sparse, but empirically inspectable.

4 REAL-DATA EMPIRICAL RESULTS

We now examine whether the fixed-prototype restriction preserves hedging quality on historical data and whether it produces an inspectable policy. All numbers are reported as normalized liability offsets, not complete economic P&L. Consequently, a value closer to zero is better for the mean and lower-tail statistics, but zero is not a break-even threshold. Comparisons are made under the same liability, hedge instruments, costs, and bounds. The numerical tables report the current panel estimates and are interpreted descriptively; the protocol-locked rerun will add the paired circular-block intervals specified in Section 3.5 before inferential claims are made.

4.1 European-Option Panel Results

Table 3 gives the equal-ticker average for the European short-call task. The two ProtoHedge rows correspond to configurations chosen by the prespecified mean- and tail-oriented validation rules. This is important: they represent two ex ante decision objectives, not two models chosen after viewing the test metric.

The first result is that the neural policies improve lower-tail offset relative to the transparent baselines. Deep Hedging increases average CVaR_{5%} from -0.137235 under Spot-Delta to -0.134793 . ProtoHedge improves it further: the Validation-Mean policy reaches -0.133763 , while the Validation-Tail policy reaches -0.132121 . Thus the interpretation constraint does not produce a visible panel-level tail penalty in this experiment.

The mean-oriented ProtoHedge policy also has the highest average offset. Its improvement over Deep Hedging is 0.000625 , or approximately 2.07% relative to the magnitude of the Deep Hedging mean offset. The tail-oriented policy sacrifices some of that mean advantage but remains 0.000132 above Deep Hedging while improving CVaR_{5%} by 0.002672 , approximately 1.98% of the Deep Hedging tail-loss magnitude. The two selections therefore expose a small but economically coherent mean–tail tradeoff rather than a universal winner.

Table 4 checks whether these averages are driven by a small subset of assets. The Validation-Mean policy has a higher mean offset than Deep Hedging for all ten tickers. The Validation-Tail policy has a higher CVaR_{5%} for all ten. The absolute improvements are largest for the most volatile single-name series, NVDA, but the direction also holds for broad market, sector, and Treasury ETFs.

These differences are modest, which is consistent with the objective of the paper. ProtoHedge is not intended to obtain large gains by solving a different hedging problem; it is intended to retain the quality of a black-box policy while exposing a finite set of representative states. The relevant empirical finding is therefore that the prototype bottleneck remains competitive across the panel, not that it creates a new source of arbitrage.

4.2 Position-Bound Analysis

The performance result is more credible if it is not generated by persistent saturation at the hard inventory limits. In the European task, average bound occupancy is 0.0775 for Deep Hedging, 0.0486 for Validation-Mean ProtoHedge, and 0.0304 for Validation-Tail ProtoHedge. These correspond to average reductions of approximately 37.3% and 60.7%, respectively. The prototype policies therefore achieve their panel-level mean and tail results while operating more often in the interior of the feasible set.

The ticker-level pattern is not uniform. The tail-oriented ProtoHedge policy has lower occupancy than Deep Hedging for six of ten underlyings, including large reductions for AAPL, IWM, NVDA, QQQ, and SPY. It has higher occupancy for TLT, XLE, XLF, and XLV, where the Deep Hedging benchmark is at zero or close to it. This heterogeneity is why occupancy is reported beside performance rather than used as a stand-alone measure of model quality. A lower-bound policy is not automatically a better hedge; it is favorable only when offset quality is preserved.

The Asian task produces a different boundary pattern, discussed below. This cross-liability contrast is informative: the prototype

Model	Mean offset	Offset CVaR _{5%}	Bound occupancy
Unhedged	-0.030502	-0.140291	0.000000
Spot-Delta	-0.030331	-0.137235	0.000000
Spot-Delta Band	-0.030295	-0.137247	0.000000
Deep Hedging	-0.030277	-0.134793	0.077515
ProtoHedge (Validation-Mean)	-0.029651	-0.133763	0.048616
ProtoHedge (Validation-Tail)	-0.030145	-0.132121	0.030434

Table 3: European short-call results, averaged across seeds within ticker and then equally across ten tickers. Higher mean offset and CVaR_{5%} are better. Bound occupancy is the fraction of path–time–instrument inventories at a hard cumulative-position limit; lower is better among learned policies.

Ticker	Mean offset		CVaR _{5%}		Bound occupancy	
	DH	Proto-Mean	DH	Proto-Tail	DH	Proto-Tail
AAPL	-0.0396	-0.0379	-0.1479	-0.1464	0.1250	0.0096
IWM	-0.0271	-0.0270	-0.1240	-0.1225	0.0907	0.0028
NVDA	-0.0724	-0.0699	-0.3265	-0.3204	0.1744	0.0116
QQQ	-0.0288	-0.0286	-0.1104	-0.1069	0.1660	0.0304
SPY	-0.0217	-0.0216	-0.0859	-0.0840	0.1927	0.0001
TLT	-0.0143	-0.0134	-0.0819	-0.0811	0.0000	0.0835
XLE	-0.0286	-0.0285	-0.1705	-0.1688	0.0000	0.0151
XLF	-0.0238	-0.0237	-0.1048	-0.1021	0.0000	0.1361
XLK	-0.0281	-0.0277	-0.1093	-0.1064	0.0264	0.0119
XLV	-0.0184	-0.0182	-0.0866	-0.0827	0.0000	0.0032

Table 4: Ticker-level European comparison. Proto-Mean is the validation-mean selection and Proto-Tail is the validation-tail selection. The paired columns deliberately show the criterion targeted by each selector. Higher offsets and lower occupancy are preferable.

bottleneck appears to regularize inventory in the European experiment, but path dependence can require a different use of the feasible action region.

4.3 Interpretability Verification: QQQ Case Study

Interpretability requires evidence about the prototype mechanism itself, not only competitive performance. For a detailed case study, we examine policies for the Invesco QQQ Trust (QQQ), a highly traded exchange-traded fund that tracks the Nasdaq-100. QQQ provides a long and liquid option history and has complete representative artifacts for both liabilities, allowing the effect of path dependence on prototype use to be studied on the same underlying. It is used for interpretability illustration, not because it had the strongest test performance; panel-level performance claims continue to use all available tickers. Table 5 reports cumulative prototype-assignment mass and two effective-library-size statistics. The entropy effective size is

$$K_{\text{ent}} = \exp\left(-\sum_{k=1}^K \bar{\alpha}_k \log \bar{\alpha}_k\right),$$

and the inverse-concentration size is

$$K_{\text{inv}} = \left(\sum_{k=1}^K \bar{\alpha}_k^2\right)^{-1},$$

where $\bar{\alpha}_k$ is the prototype’s mean assignment weight over test paths and decision times. Both equal K under perfectly uniform use and approach one as assignment mass concentrates.

The European model uses a compact effective dictionary. Three of its 25 prototypes explain 58.4% of assignment mass, ten explain 99.6%, and its effective size is between seven and eight. The prototype layer is therefore not a decorative uniform mixture: most decisions pass through a substantially smaller set of recurring states. The Asian model uses more of its available library. Its top ten prototypes explain 71.3% and its effective size is between 13 and 19. This is still concentrated relative to 50 uniform prototypes, but it suggests that a path-dependent payoff requires a richer set of state representatives.

Table 6 reports the most heavily weighted prototype within return, volatility, and drawdown terciles. In the European model, prototype 21 dominates down-return, high-volatility, and mild-drawdown states, whereas prototype 8 dominates middle/up returns, low/middle volatility, and middle/deep drawdowns. This systematic switch is consistent with regime-aware example use. In the Asian model, prototype 10 remains the single most-used anchor in every tercile, but its weight falls from 0.215 in low volatility to 0.182 in high volatility while the remaining assignment mass redistributes. The Asian evidence therefore supports state-dependent weighting, but not the stronger claim that the identity of the top prototype changes in every regime.

Liability	Representative policy	K	Top 1	Top 3	Top 5	Top 10	K_{ent}	K_{inv}
European	QQQ ProtoHedge	25	0.208	0.584	0.747	0.996	8.29	7.04
Asian	QQQ ProtoHedge	50	0.199	0.352	0.480	0.713	18.78	13.04

Table 5: Concentration of prototype assignment mass for the representative QQQ case study. Top- k columns report cumulative mass after sorting prototypes by mean use. Effective sizes below the nominal K indicate non-uniform use.

Liability	Regime partition	Low / Down / Mild	Middle	High / Up / Deep
European	Return	P21 (0.206)	P8 (0.250)	P8 (0.242)
European	Volatility	P8 (0.313)	P8 (0.203)	P21 (0.208)
European	Drawdown	P21 (0.210)	P8 (0.225)	P8 (0.295)
Asian	Return	P10 (0.199)	P10 (0.204)	P10 (0.193)
Asian	Volatility	P10 (0.215)	P10 (0.200)	P10 (0.182)
Asian	Drawdown	P10 (0.190)	P10 (0.200)	P10 (0.206)

Table 6: Dominant prototype and mean assignment weight in the QQQ case study. Regimes are test-period terciles defined separately for terminal return, average implied volatility, and maximum drawdown.

Nearest-state retrieval supplies the final link from model internals to market states. Table 7 reports the closest observed QQQ state for four highly used European prototypes. For example, P8 represents a state with normalized spot above inception and a moderate positive spot inventory, whereas P21 is associated with normalized spot near 0.85, a more expensive listed call, and a larger spot inventory. P11 and P16 occupy intermediate price and inventory regions. These are concrete observed states, not latent centroids. The retrieved path identifiers allow the complete 20-step histories to be inspected and, in the locked rerun, mapped to calendar dates.

Together, concentration, regime use, and nearest-state retrieval verify three different aspects of interpretability: the policy uses a finite and non-uniform state dictionary, the assignment mechanism responds to market conditions, and dominant anchors can be inspected as observed states. They do not yet establish that professional hedgers find the explanations operationally useful, nor do they justify labeling a retrieved state as a crisis or flash crash without its calendar metadata.

4.4 Asian-Option Results

The Asian experiment changes the liability and adds running-average moneyiness to every learned policy and prototype state, while preserving the data split, hedge instruments, costs, candidate grid, and selection rules. It therefore tests whether ProtoHedge remains competitive when current prices alone are not a sufficient liability state.

The principal performance conclusion survives path dependence. Relative to Deep Hedging, the mean-selected policy improves average mean offset by 0.000271 and CVaR_{5%} by 0.000541. It wins seven of nine available ticker comparisons on each metric. The tail-selected policy has essentially the same mean offset as Deep Hedging, a difference of 0.000015, while improving average CVaR_{5%} by 0.002883, or approximately 3.88% of the Deep Hedging tail-loss magnitude. Its CVaR is higher for all nine available tickers. This is the strongest current evidence that the prototype bottleneck remains useful when the liability is path-dependent.

The position result does not carry over unchanged. The mean-selected policy has average occupancy 0.0196 versus 0.0185 for Deep Hedging, and the tail-selected policy reaches 0.0382. Unlike the European task, neither Asian selection lowers occupancy ticker by ticker. The path-dependent liability can therefore preserve the performance–interpretability tradeoff without preserving the same inventory-regularization pattern. This distinction is more informative than a blanket claim that ProtoHedge always trades less aggressively.

Table 10 summarizes the central comparison with Deep Hedging. Across both liabilities, the validation-tail policy provides the most consistent lower-tail improvement. The mean selector provides the larger average-offset gain. The European and Asian occupancy columns then show why performance and constraint use must be assessed jointly rather than collapsed into one score.

Overall, the Asian experiment extends rather than merely repeats the European finding. The learned policy receives the payoff state that makes the Asian problem Markov in the supplied information, the tail-oriented prototype policy remains competitive on every available ticker, and its prototype library is inspectable but less concentrated. At the same time, higher boundary use shows that path dependence changes the economic behavior of the selected policy.

5 DISCUSSION AND LIMITATIONS

The results answer the paper’s central empirical question in qualified but favorable terms. A fixed prototype bottleneck can remain competitive with a common-protocol Deep Hedging network on historical data. In the European task, the mean-oriented and tail-oriented selectors improve the corresponding panel averages and reduce average use of hard position limits. In the Asian task, the same selectors preserve mean and lower-tail competitiveness after adding the running payoff state, but the reduction in boundary use disappears. The interpretability evidence also changes with the liability: the European policy uses a compact effective dictionary and switches its dominant anchor across several regimes, whereas the

Prototype	Path	Step	Spot position	Option position	Norm. spot	Norm. option
P8	315	11	0.5612	-0.0012	1.0464	0.0183
P11	497	5	0.4701	-0.0013	0.9940	0.0600
P16	36	5	0.5232	-0.0011	1.0129	0.0255
P21	372	7	0.6582	-0.0018	0.8492	0.0673

Table 7: Nearest observed states in the QQQ case study for four dominant European prototypes. Prices are normalized by episode-initial spot. These state summaries support example-based inspection; calendar-event labels will be added from the contract-consistent rerun rather than inferred from path shape alone.

Model	Mean offset	Offset CVaR _{5%}	Bound occupancy	Tickers
Unhedged	-0.016479	-0.078681	0.000000	10
Spot-Delta	-0.016265	-0.074999	0.000000	10
Spot-Delta Band	-0.016240	-0.074893	0.000000	10
Deep Hedging	-0.015801	-0.074384	0.018519	9
ProtoHedge (Validation-Mean)	-0.015530	-0.073844	0.019606	9
ProtoHedge (Validation-Tail)	-0.015786	-0.071501	0.038187	9

Table 8: Asian short-call results averaged across the available ticker runs. The final column reports the number of tickers contributing to each row.

Ticker	Mean offset		CVaR _{5%}		Bound occupancy	
	DH	Proto-Mean	DH	Proto-Tail	DH	Proto-Tail
IWM	-0.0154	-0.0151	-0.0751	-0.0715	0.0000	0.0000
NVDA	-0.0375	-0.0364	-0.1672	-0.1635	0.1667	0.1949
QQQ	-0.0154	-0.0156	-0.0627	-0.0583	0.0000	0.0000
SPY	-0.0120	-0.0120	-0.0532	-0.0519	0.0000	0.0573
TLT	-0.0080	-0.0073	-0.0449	-0.0427	0.0000	0.0068
XLE	-0.0154	-0.0153	-0.0882	-0.0855	0.0000	0.0814
XLF	-0.0133	-0.0132	-0.0643	-0.0616	0.0000	0.0014
XLK	-0.0149	-0.0148	-0.0618	-0.0591	0.0000	0.0007
XLV	-0.0103	-0.0101	-0.0522	-0.0493	0.0000	0.0012

Table 9: Ticker-level Asian comparison across the available learned-policy runs.

Liability	Selection	N	Δ Mean	Mean wins	Δ CVaR	CVaR wins	Δ Bound
European	Validation-Mean	10	0.000625	10	0.001030	7	-0.028898
European	Validation-Tail	10	0.000132	6	0.002672	10	-0.047081
Asian	Validation-Mean	9	0.000271	7	0.000541	7	0.001087
Asian	Validation-Tail	9	0.000015	5	0.002883	9	0.019669

Table 10: ProtoHedge minus Deep Hedging, averaged equally across available tickers. Positive mean and CVaR differences favor ProtoHedge; negative bound differences favor ProtoHedge. “Wins” counts tickers with a favorable paired point estimate. Statistical claims require the locked block-bootstrap intervals.

Asian policy distributes mass across a larger library while retaining one dominant prototype.

5.1 Economic and Modeling Interpretation

The magnitude of the performance differences should be interpreted in the context of the research objective. All learned models trade the same two instruments, observe the same task-relevant information, optimize the same monetary utility, and face the same costs and

bounds. ProtoHedge’s contribution is therefore not a fundamentally different source of hedging returns. It is a restriction on how a neural action is represented. Preserving or slightly improving offset quality under that restriction is evidence that the black-box benchmark may use more representational freedom than this empirical task requires.

The separate validation selectors are also informative. The mean-oriented configuration yields the larger average-offset improvement, while the tail-oriented configuration yields the more consistent CVaR improvement. This means that “best ProtoHedge” is not a single economically neutral concept. The appropriate model depends on whether a decision maker prioritizes average liability offset or severe-loss mitigation. The prototype layer makes this choice easier to audit because the selected policy can be examined through its prototype actions, assignment concentration, and representative states.

Position-bound occupancy adds a behavioral dimension that is invisible in mean and CVaR alone. Lower European occupancy suggests that the prototype restriction can act as an inventory regularizer. The Asian result prevents us from generalizing that claim: path-dependent hedging uses the feasible region more aggressively even when performance remains competitive. The practical lesson is that interpretability, tail protection, and conservative inventory use are related but distinct properties. They should be reported jointly, not combined into a single ranking.

5.2 Limitations and Cases Without Dominance

ProtoHedge does not dominate every model, asset, or criterion. The mean-oriented and tail-oriented policies are different validation selections; neither is claimed to optimize all metrics simultaneously. European occupancy is higher than Deep Hedging for several individual tickers even though the panel average is lower, and both Asian selections have higher average occupancy. Some per-ticker mean comparisons also favor Deep Hedging. These exceptions are part of the empirical result rather than failures to be hidden: they identify settings in which the prototype restriction or the selected objective trades flexibility for auditability.

Several broader limitations remain. First, the reported outcome excludes the initial liability premium. It is a liability-offset measure for controlled policy comparison, not a complete terminal P&L, and zero does not represent economic break-even. A complete P&L study would require an observed or independently priced liability premium under a clearly stated valuation rule. This is also why the negative-offset frequency is not a headline metric.

Second, the transaction-cost formula is explicit and applied consistently but is stylized. The backtest does not reconstruct executable bid and ask trades, market impact, funding, margin, assignment risk, or desk-level risk limits. It also assumes that the selected listed hedge option remains tradable at the observed daily midpoint throughout an episode. Contract continuity corrects the self-financing identity, but it does not eliminate execution uncertainty.

Third, the panel is broad relative to a single-ticker case study but remains limited to liquid U.S. equities and ETFs. It does not establish external validity for index options, foreign exchange, commodities, rates, digital assets, or less liquid single names. The 20-step horizon and near-ATM call selection similarly cover only one part of the option surface.

Fourth, windows overlap within each calendar period. Splitting before window construction and imposing a 19-observation embargo eliminate shared dates across train, validation, and test sets, but they do not make adjacent test windows independent. The final

claims must therefore use the paired circular-block intervals specified in Section 3.5, not standard errors that treat all episodes as independent. The current point estimates should not be described as statistically significant until those intervals are populated.

Fifth, the final protocol-locked run is required to regenerate every table under cumulative inventory accounting, chronological purged splits, fixed hedge contracts, payoff-sufficient Asian state, and validation-only configuration selection. The paper is structured so those results replace numerical cells without changing definitions or claims logic.

Sixth, inspectability is not the same as practitioner validation. Concentrated weights and nearest historical states show what the model uses, but the present study does not ask professional hedgers whether those explanations improve model review or trading decisions. Likewise, the retrieved-state artifact does not yet attach calendar dates and event annotations. We therefore do not claim that the current prototypes have already been verified as financial-crisis or flash-crash states. Date-aware retrieval and a prespecified event study are needed for that stronger claim.

Finally, the benchmark set is intentionally focused: transparent delta rules and a common-protocol feed-forward Deep Hedging network. It does not include the full range of recurrent, transformer, distributional, or dynamics-aware deep hedging methods, including Deeper Hedging under Chiarella–Heston dynamics [3]. Nor does the paper complete the original scalability agenda. The grid varies K , source policy, and similarity weighting, but it does not systematically isolate wall-clock complexity, memory use, feature dimension, or approximate nearest-neighbor search. Those are distinct comparative and systems studies rather than evidence already supplied by this run.

5.3 Relation to the ProtoHedge Research Agenda

The original ProtoHedge paper identified empirical validation, path-dependent instruments, historical-scenario interpretation, scalability, and broader model comparison as future work [2]. The present study directly advances the first two items: it constructs a real-data panel, enforces historical out-of-sample evaluation, and adds an Asian liability with a payoff-sufficient state. It partially advances historical interpretation by measuring concentration, regime use, and nearest-state retrieval. It does not claim to complete the professional-user, scalability, or broad-benchmark components.

This distinction keeps the paper’s contribution clear. A single additional run cannot credibly “solve” every future-work axis because each requires a different experimental design. Scalability requires controlled timing and memory experiments as dimensionality and K vary. Comparative evaluation requires faithful implementation and tuning of additional models. Practitioner validation requires a human-subject or structured expert protocol. These are natural follow-on projects, but they should not dilute the central contribution of the present empirical test.

6 CONCLUSION

This paper tests whether ProtoHedge’s fixed, example-based representation survives the transition from simulation to historical option data. The framework uses one consistent listed hedge contract

per episode, chronological purged splits, and cumulative-inventory accounting. It also imposes stylized costs, hard action and position bounds, validation-only selection, and a payoff-sufficient state for the Asian liability. The reported outcome excludes the liability premium and therefore is not complete economic P&L.

Across the current panel estimates, ProtoHedge remains competitive with Deep Hedging on both liabilities. The European mean and tail selections improve their targeted average offset metrics while reducing average bound occupancy. The Asian selections preserve the performance result, particularly in the lower tail, but use the hard bounds more frequently. Prototype assignments are non-uniform, respond to market regimes, and map back to observed historical states; the Asian policy uses a larger effective dictionary than the European policy. These findings support the original ProtoHedge thesis in a qualified empirical form: interpretability can be introduced at little apparent cost to hedging quality, but its effects on inventory behavior and prototype concentration depend on the liability.

The remaining work before submission is finite and explicit: complete the protocol-locked rerun, populate paired circular-block confidence intervals, attach dates to nearest historical states, and update the numerical cells without changing the prespecified analysis. Subject to those checks, the paper provides the first systematic real-data validation of the fixed-prototype hedging framework and a foundation for subsequent work on scalable comparison and trainable prototype locations.

ETHICS AND PRIVACY STATEMENT

This study uses licensed historical market data and does not involve human subjects or personal data. A principal risk is that backtest results could be misinterpreted as evidence of live-trading profitability; we therefore state the premium convention, execution assumptions, uncertainty, and external validity limitations explicitly. Prototype explanations may improve model auditability, but they should support rather than replace professional risk management, independent validation, and applicable regulatory controls.

REFERENCES

- [1] Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. 2019. Deep Hedging. *Quantitative Finance* 19, 8 (2019), 1271–1291. <https://doi.org/10.1080/14697688.2019.1571683>
- [2] Lisa Faloughi, Ce Guo, and Wayne Luk. 2025. ProtoHedge: Interpretable Hedging with Market Prototypes. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. Association for Computing Machinery, 202–210. <https://doi.org/10.1145/3768292.3770347>
- [3] Kang Gao, Stephen Weston, Perukrishnen Vytelingum, Namid Stillman, Wayne Luk, and Ce Guo. 2023. Deeper Hedging under Chiarella–Heston Dynamics. In *Proceedings of the 4th ACM International Conference on AI in Finance (ICAIF '23)*. Association for Computing Machinery, 270–278. <https://doi.org/10.1145/3604237.3626913>
- [4] James MacQueen. 1967. Some Methods for Classification and Analysis of Multivariate Observations. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, Vol. 1. University of California Press, 281–297.
- [5] Dimitris N. Politis and Joseph P. Romano. 1992. A Circular Block-Resampling Procedure for Stationary Data. In *Exploring the Limits of Bootstrap*, Raoul LePage and Lynne Billard (Eds.). Wiley, 263–270.
- [6] R. Tyrrell Rockafellar and Stanislav Uryasev. 2000. Optimization of Conditional Value-at-Risk. *The Journal of Risk* 2, 3 (2000), 21–41. <https://doi.org/10.21314/JOR.2000.038>
- [7] A. Elizabeth Whalley and Paul Wilmott. 1997. An Asymptotic Analysis of an Optimal Hedging Model for Option Pricing with Transaction Costs. *Mathematical Finance* 7, 3 (1997), 307–324. <https://doi.org/10.1111/1467-9965.00034>

A OUTCOME AND INFERENCE DETAILS

For a set of test offsets $\{O_i\}_{i=1}^n$, let $q_{0.05}$ be the empirical fifth percentile. The reported lower-tail statistic is

$$\widehat{\text{CVaR}}_{5\%} = \frac{1}{|\mathcal{I}_{0.05}|} \sum_{i \in \mathcal{I}_{0.05}} O_i, \quad \mathcal{I}_{0.05} = \{i : O_i \leq q_{0.05}\}.$$

Because the outcome is a gain-like offset, a larger value is preferable. For RMSE and downside deviation, the comparison signs are reversed before reporting a ProtoHedge improvement so that every paired effect has the same orientation.

For each ticker, liability, selector, and test episode, offsets are first averaged across available training seeds. ProtoHedge and Deep Hedging are then kept paired on the same episode indices. A circular block sample concatenates random blocks of 20 adjacent episode indices, wrapping at the end of the test sequence, until the original sample length is reached. The paired statistic is recomputed for each of 2,000 replicates. Panel replicates resample within every ticker independently and then take the equal-ticker average. Percentile 2.5% and 97.5% quantiles define the reported interval.

B DATA-INTEGRITY CHECKS

The final data artifact records the following checks for every retained sample:

- (1) all 20 observations belong to one chronological split and one contiguous date block;
- (2) the 19-observation embargo is applied after each split boundary;
- (3) one optionid, strike, and expiration are used throughout the episode;
- (4) the listed hedge option remains unexpired after the liability horizon and passes the initial liquidity and moneyness filters;
- (5) one-step listed-option returns are multiplied by cumulative option inventory, not only the newest trade;
- (6) Asian policies and prototype features contain running average moneyness at every decision time; and
- (7) model configuration files identify whether an artifact was produced by the corrected scientific protocol, preventing reuse of superseded random-split or contract-switching results.

C RECOMMENDED SUPPLEMENTARY REPORTING

The main paper emphasizes compact tables. The accompanying supplementary artifact should archive, without adding visual clutter to the core argument:

- (1) full per-ticker results for every classical baseline, Deep Hedging, and both validation-selected ProtoHedge policies;
- (2) validation-only candidate tables over prototype count, source policy, and similarity weighting;
- (3) per-ticker paired circular-block intervals for mean offset, $\widehat{\text{CVaR}}_{5\%}$, RMSE, and downside deviation;
- (4) prototype actions, assignment frequencies, concentration curves, and regime-conditioned assignment tables for each selected model; and

- (5) nearest historical states with ticker, date, contract identifier, strike, expiration, payoff state, and complete 20-step price trajectory.

These materials separate model selection from final test reporting and permit readers to inspect the evidence underlying every aggregate claim.